

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 3.2 PROBABILITY I – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can Kamau, and other Kenyans facing uncertain everyday outcomes, use the mathematics of probability to make smarter, fairer, and safer decisions? Kamau is a matatu conductor at Machakos Country Bus Station. His 14-seater matatu runs the Nairobi–Kitengela route. His 20-trip tally notebook records these passenger counts: 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14. A trip with 14 passengers is a fully booked (full) trip. Answer every section in the space under each prompt. Show your working, write fractions as well as decimals where asked, and use full sentences when you are asked to explain. Refer to Kamau's data and to what you learned in Lessons 1 to 8.

Section 1: Why can't Kamau predict one trip, yet still say something useful?

(a) Use Kamau's 20 trip counts to find the experimental probability that a trip is fully booked. Show how you counted. (b) In Lesson 1 you tossed a 10-shilling coin 30 times in pairs. No pair got exactly 15 heads, but the pooled class total was close to 0.5. Explain what this tells you about experimental probability and the number of trials. (c) In your own words, explain why Kamau cannot know for certain whether his next trip will be full, yet can still make a useful mathematical statement about his chances.

Write your answer here:

Section 2: The probability scale and the sample space

(a) Kamau's matatu stops at Mlolongo and then Athi River. At each stop a particular seat is either Taken (T) or Empty (E). List the complete sample space for the two stops, writing the outcomes as ordered pairs. How do you know your list is complete and has no repeats? (b) Kamau's $P(\text{Full}) = 0.4$ and $P(\text{Not Full}) = 0.6$. Place these on a 0 to 1 probability scale in words, and explain why no probability from his notebook could ever be -0.3 or 1.4 . (c) Give one example of an impossible event and one certain event from Kamau's matatu world, with their probabilities.

Write your answer here:

Section 3: Mutually exclusive events and the Addition Law

Let F be the event 'a trip is fully booked (14 passengers)' and L be the event 'a trip has fewer than 7 passengers'. (a) Use Kamau's 20 trip counts to find $P(F)$ and $P(L)$ as fractions. (b) Are F and L mutually exclusive? Explain. (c) Find $P(F \text{ or } L)$ using the Addition Law, and say what it means for Kamau in a sentence.

Write your answer here:

Section 4: Independent events, the Multiplication Law and a tree diagram

Kamau runs a morning and an afternoon trip. Use $P(\text{Full}) = 0.4$ for each trip, and assume the two trips are independent. (a) What does it mean for the two trips to be independent? (b) Draw or describe a two-stage tree diagram with all four end outcomes and write the probability of each. (c) Find $P(\text{full on both trips})$ and $P(\text{full on exactly one trip})$. (d) Show how you can check that your tree diagram probabilities are correct.

Write your answer here:

Section 5: Advising Kamau – the Probability Report

Kamau's employer asks for advice. (a) Write a short Probability Report (4 to 6 sentences) saying what Kamau should expect about full buses, how he should plan, and what he cannot know. Use numbers from his notebook. (b) Suppose each of the 14 seats was filled independently with probability $10/14$. The probability that all 14 seats are filled would then be about 0.009, but Kamau's data gave 0.4. What does this mismatch tell you about the independence assumption and about the model? (c) Finish this sentence in your own words: 'Probability helps Kenyans make smarter, fairer and safer decisions because...' Give one example from the unit, such as a fair game or a road-safety decision.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Experimental probability and use of data	Correctly counts 8 full trips from the tally, gets $8/20 = 0.4$, and clearly links more trials to a more reliable estimate.	Gets 0.4 correctly but gives only a brief explanation of why more trials matter.	Miscounts the data or does not use the relative-frequency definition.
Sample space, probability scale and the Addition Law	Lists {TT, TE, ET, EE} correctly, explains why $0 \leq P \leq 1$, and correctly applies the Addition Law with a justification of mutual exclusivity.	Gets the sample space and the answer of $1/2$ but gives a weak justification for the scale or mutual exclusivity.	Omits outcomes, gives a probability outside 0 to 1, or adds probabilities without checking the events.
Independent events, Multiplication Law and tree diagram	Defines independence accurately, completes all four branches with correct products (0.16, 0.24, 0.24, 0.36), finds 0.16 and 0.48, and uses the sum-to-1 check.	Gets most branch probabilities and the main answers right but misses the check or explains independence loosely.	Adds instead of multiplies along branches, or leaves out outcomes.
Reasoning, model critique and communication	Writes a clear report with numbers, explains why the independent-seats model fails, and links probability to fair or safe decisions in Kenyan contexts.	Gives usable advice with numbers but only a partial explanation of the model mismatch.	Gives vague advice with no numbers or no link to the data.

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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